

QUESTIONS

Q1

Smart Document Editor using
String, StringBuilder and
StringBuffer
10 marks

Q2

Hospital Token Management
System using List Interface
10 marks

Q3

Airport Boarding Queue
Management using Queue
Interface
10 marks

Q4

Digital Certificate Verification
using Set Interface
10 marks

Q5

Smart Inventory Management
using Map Interface
10 marks

5/5 answered

Q1 Smart Document Editor using String, StringBuilder and StringBuffer

10 marks

Question 1 :

- 1.Read a paragraph from the user.
 - 2.Count the number of characters, words, and sentences.
 - 3.Replace every occurrence of a given word with another word.
 - 4.Reverse each sentence individually.
 - 5.Maintain editing history using StringBuffer.
 - 6.Construct the final formatted document using StringBuilder.
 - 7.Display the original document, editing history, and final document.
- Requirement: Use String methods wherever appropriate. Use StringBuilder for document construction and StringBuffer for edit history.

Your Code

```
1 import java.util.Scanner;  
2  
3 public class SmartDocumentEditor {  
4     public static void main(String[] args) {  
5         Scanner scanner = new Scanner(System.in);  
6  
7         System.out.println("Enter the paragraph:");  
8         String paragraph = scanner.nextLine();  
9  
10        System.out.println("Enter the word to replace:");  
11        String targetWord = scanner.nextLine();  
12    }  
13 }
```

Input

Enter the paragraph:
Artificial Intelligence is
transforming education. Students use
AI for learning.

Output

Enter the paragraph:
Enter the word to replace:
Enter the replacement word:

--- ORIGINAL DOCUMENT ---

10 marks

Q4

Digital Certificate Verification
using Set Interface
10 marks

Q5

Smart Inventory Management
using Map Interface
10 marks

5/5 answered

Your Code

```
1 import java.util.Scanner;
2
3 public class SmartDocumentEditor {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         System.out.println("Enter the paragraph:");
8         String paragraph = scanner.nextLine();
9
10        System.out.println("Enter the word to replace:");
11        String targetWord = scanner.nextLine();
12
13        System.out.println("Enter the replacement word:");
14        String replacementWord = scanner.nextLine();
15
16        StringBuffer history = new StringBuffer();
17        history.append("History Log:\n");
18        history.append("- Document created.\n");
19
20        int charCount = paragraph.length();
21        int wordCount = paragraph.trim().isEmpty() ? 0 : paragraph.trim().split(" ").length;
22        int sentenceCount = paragraph.trim().isEmpty() ? 0 : paragraph.trim().split("\\.")[0].length;
23
24        String updatedParagraph = paragraph.replace(targetWord, replacementWord);
25        history.append("- Replaced word ' " + targetWord + " ' with ' " + replacementWord + " '");
26        history.append("\n");
27        history.append(updatedParagraph);
28    }
}
```

Input

Enter the paragraph:
Artificial Intelligence is
transforming education. Students use
AI for learning.

Output

Character Count : 20
Word Count : 3
Sentence Count : 1

--- EDITING HISTORY ---
History Log:
- Document created.

```
30
31     StringBuilder finalDocument = new StringBuilder();
32     String[] sentences = updatedParagraph.split("(?<=[.!?])");
33
34     for (String sentence : sentences) {
35         if (!sentence.trim().isEmpty()) {
36             StringBuilder sb = new StringBuilder(sentence.trim());
37             finalDocument.append(sb.reverse()).append("\n");
38         }
39     }
40
41     history.append("- Reversed each sentence individually.\n");
42
43     System.out.println("\n--- ORIGINAL DOCUMENT ---");
44     System.out.println(paragraph);
45
46     System.out.println("\n--- DOCUMENT STATISTICS ---");
47     System.out.println("Character Count : " + charCount);
48     System.out.println("Word Count : " + wordCount);
49     System.out.println("Sentence Count : " + sentenceCount);
50
51     System.out.println("\n--- EDITING HISTORY ---");
52     System.out.println(history);
53
54     System.out.println("--- FINAL DOCUMENT ---");
55     System.out.println(finalDocument);
56
57     scanner.close();
58 }
```

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Digital Certificate Verification
using Set Interface
10 marks**Q5** ✓
Smart Inventory Management
using Map Interface
10 marks**Q2 Hospital Token Management System using List Interface**

10 marks

Hospital Token Management System using List Interface

Store patient details using an appropriate List implementation.

Register new patients.

Display patients in registration order.

Update patient information using Patient ID.

Remove cancelled appointments.

Display all patients of a particular department.

Find the first and last registered patients.

Requirement: Use a suitable List implementation

Your Code

```
1 import java.util.*;  
2  
3 class Main {  
4     public static void main(String[] args) {  
5         Scanner sc = new Scanner(System.in);  
6         List<String> list = new ArrayList<>();
```

Input

Enter number of patients:
3

Enter ID Name Department:

Output

```

1 import java.util.*;
2
3 class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         List<String> list = new ArrayList<>();
7         list.add("101 Ram Cardiology");
8         list.add("102 Sita Neurology");
9         list.add("103 Ravi Cardiology");
10
11        System.out.println("Patients:");
12        for (String s : list)
13            System.out.println(s);
14
15        for (int i = 0; i < list.size(); i++)
16            if (list.get(i).startsWith("102"))
17                list.set(i, "102 Anita Neurology");
18
19        list.removeIf(s -> s.startsWith("101"));
20
21        System.out.println("\nCardiology Patients:");
22        for (String s : list)
23            if (s.contains("Cardiology"))
24                System.out.println(s);
25
26        System.out.println("\nFirst: " + list.get(0));
27        System.out.println("Last: " + list.get(list.size() - 1));
28    }

```

Enter number of patients:
3

Enter ID Name Department:

Output

Patients:
101 Ram Cardiology
102 Sita Neurology
103 Ravi Cardiology

Cardiology Patients:

10 marks

Q4

Digital Certificate Verification
using Set Interface
10 marks

Q5

Smart Inventory Management
using Map Interface
10 marks

5/5 answered

Requirement: Use appropriate Queue implementations.

Your Code

```
1 import java.util.Scanner;  
2 import java.util.Queue;  
3 import java.util.LinkedList;  
4 import java.util.PriorityQueue;  
5  
6 public class Main {  
7     public static void main(String[] args) {  
8  
9         Scanner sc = new Scanner(System.in);  
10  
11         Queue<String> queue = new LinkedList<>();  
12         PriorityQueue<String> vip = new PriorityQueue<>();  
13  
14         System.out.print("Enter number of passengers: ");  
15         int n = Integer.parseInt(sc.nextLine());  
16  
17         System.out.println("Enter passenger names:");  
18         for (int i = 0; i < n; i++) {  
19             queue.offer(sc.nextLine());  
20         }  
21  
22         System.out.print("Enter number of VIP passengers: ");  
23         int m = Integer.parseInt(sc.nextLine());  
24  
25         System.out.println("Enter VIP passenger names:");
```

Input

Ravi
2
Arun
Kiran

Output

Enter VIP passenger names:
Current Queue: [Ram, Sita, Ravi]
Next Passenger: Ram
VIP Boarded: Arun
Remaining VIP Queue: [Kiran]
VIP Boarded: Kiran



```
16
17     System.out.println("Enter passenger names:");
18     for (int i = 0; i < n; i++) {
19         queue.offer(sc.nextLine());
20     }
21
22     System.out.print("Enter number of VIP passengers: ");
23     int m = Integer.parseInt(sc.nextLine());
24
25     System.out.println("Enter VIP passenger names:");
26     for (int i = 0; i < m; i++) {
27         vip.offer(sc.nextLine());
28     }
29
30     System.out.println("Current Queue: " + queue);
31
32     if (!queue.isEmpty()) {
33         System.out.println("Next Passenger: " + queue.peek());
34     }
35
36     while (!vip.isEmpty()) {
37         System.out.println("VIP Boarded: " + vip.poll());
38         System.out.println("Remaining VIP Queue: " + vip);
39     }
40
41     while (!queue.isEmpty()) {
42         System.out.println("Boarded: " + queue.poll());
43         System.out.println("Remaining Queue: " + queue);
44     }
45
46     sc.close();
47
48 }
```

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using Set Interface
10 marks**Q5**
Smart Inventory Management
using Map Interface
10 marks

5/5 answered

Q4 Digital Certificate Verification using Set Interface

10 marks

Digital Certificate Verification using Set Interface

Store certificate IDs using an appropriate Set implementation.

Prevent duplicate certificate entries.

Verify whether a certificate ID exists.

Display certificates in insertion order.

Display certificates in sorted order.

Find the total number of unique certificates issued.

Remove revoked certificates.

Requirement: Use suitable Set implementations.

Your Code

```
1 import java.util.LinkedHashSet;  
2 import java.util.Set;  
3 import java.util.TreeSet;  
4  
5 public class Main {  
6     public static void main(String[] args) {  
7  
8         Set<String> certificates = new LinkedHashSet<>();
```

Input

```
Enter number of certificates: 5  
Enter Certificate IDs:  
cert101  
cert102  
...
```

Output

```
Certificates in Insertion Order:  
cert101  
cert102
```


Hospital Token Management
System using List Interface
10 marks

Q3
Airport Boarding Queue
Management using Queue
Interface
10 marks

Q4
Digital Certificate Verification
using Set Interface
10 marks

Q5
Smart Inventory Management
using Map Interface
10 marks

5/5 answered

Verify whether a certificate ID exists.

Display certificates in insertion order.

Display certificates in sorted order.

Find the total number of unique certificates issued.

Remove revoked certificates.

Requirement: Use suitable Set implementations.

Your Code

```
1 import java.util.LinkedHashSet;
2 import java.util.Set;
3 import java.util.TreeSet;
4
5 public class Main {
6     public static void main(String[] args) {
7
8         Set<String> certificates = new LinkedHashSet<>();
9         certificates.add("CERT-103");
10        certificates.add("CERT-101");
11        certificates.add("CERT-102");
12        certificates.add("CERT-101");
13
14        System.out.println("Certificates in Insertion Order: " + certifi
15
16        String searchId = "CERT-102";
17        System.out.println("Is " + searchId + " valid "
18            + certificates.contains(searchId));
19
20        System.out.println("\nTotal Unique Certificates: "
```

Input

Enter number of certificates: 5
Enter Certificate IDs:
cert101
cert102
...

Output

Certificates in Insertion Order:
[CERT-103, CERT-101, CERT-102]
Is CERT-102 valid true
Total Unique Certificates: 3

10 marks

Q5

Smart Inventory Management
using Map Interface
10 marks

5/5 answered

Your Code

```
1 import java.util.LinkedHashSet;
2 import java.util.Set;
3 import java.util.TreeSet;
4
5 public class Main {
6     public static void main(String[] args) {
7
8         Set<String> certificates = new LinkedHashSet<>();
9         certificates.add("CERT-103");
10        certificates.add("CERT-101");
11        certificates.add("CERT-102");
12        certificates.add("CERT-101");
13
14        System.out.println("Certificates in Insertion Order: " + certificates);
15
16        String searchId = "CERT-102";
17        System.out.println("Is " + searchId + " valid "
18            + certificates.contains(searchId));
19
20        System.out.println("\nTotal Unique Certificates: "
21            + certificates.size());
22
23        String revokedId = "CERT-103";
24        certificates.remove(revokedId);
25        System.out.println("\nAfter Revoking: " + certificates);
26
27        Set<String> sortedCertificates = new TreeSet<>(certificates);
28        System.out.println("\nCertificates in Sorted Order: "
29            + sortedCertificates);
30    }
31 }
```

Input

Enter number of certificates: 5
Enter Certificate IDs:
cert101
cert102

Output

Certificates in Insertion Order:
[CERT-103, CERT-101, CERT-102]
Is CERT-102 valid true
Total Unique Certificates: 3

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Smart Inventory Management
using Map Interface
10 marks

5/5 answered

Q5 Smart Inventory Management using Map Interface

10 marks

Smart Inventory Management using Map Interface

Store product information using an appropriate Map implementation.

Add new products.

Update product quantity.

Search a product using Product ID.

Calculate the total inventory value.

Display products category-wise.

Remove discontinued products.

Display all available products.

Requirement: Use a suitable Map implementation.

Your Code

```
1 import java.util.Scanner;  
2 import java.util.HashMap;  
3  
4 public class Main {  
5     public static void main(String[] args) {  
6  
7         Scanner sc = new Scanner(System.in);
```

Input

```
2  
101  
Laptop  
Electronics
```

Output

